FACULTY OF COMPUTING
UTM Johor Bahru

COURSE: SYSTEM ANALYSIS AND DESIGN

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PROJECT 3

BUS TICKET BOOKING SYSTEM

"BeepBeep"

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1.0 Overview of the Project

The Bus Ticket Booking System is an integrated, web-based information platform designed to modernize and streamline the process of purchasing and managing bus tickets. Buses remain one of the most heavily relied-upon modes of public transportation for daily commuting, student travel, and long-distance journeys. As consumer demand continuously expands, there is a critical expectation for bus operators to deliver services that are fast, accessible, and reliable.

This project serves as a digital transformation initiative aimed at replacing traditional, manual ticketing procedures with an automated, user-friendly electronic system. Built to bridge existing service gaps, the platform offers cross-platform accessibility, enabling customers to remotely register accounts, search route schedules, view real-time seat configurations, and complete secure online reservations through their personal internet-enabled devices.

In addition to elevating the customer experience, the system delivers operational advantages to transportation administrators. By establishing structured, centralized data management, bus operators can efficiently monitor sales metrics, allocate vehicle fleets, instantly modify schedules, and generate accurate financial and occupancy reports. Ultimately, the system bridges the gap between customer expectations and backend logistical workflows, driving long-term sustainability within the public transport sector.

2.0 Problem Statement

Despite widespread industrial technological integration, current local bus ticketing frameworks continue to struggle with a heavy reliance on outdated manual procedures and segregated paper-based systems. This dynamic gives rise to several fundamental problems affecting both consumers and transit service operators:

- **Inconvenient Physical Constraints**: Under the traditional manual framework, passengers are often forced to travel to physical ticketing counters to obtain basic travel information or finalize bookings. This geographical constraint leads to massive terminal congestion, particularly during weekends, school breaks, and major festive holidays, resulting in long waiting queues and extensive customer dissatisfaction.
- **Absence of Real-Time Data Transparency**: Information regarding trip schedules and layout availability is constrained to localized notice boards or verbal updates from staff. Without dynamic database synchronization, customers frequently encounter logistical issues where a seat is sold simultaneously at another counter, causing frustrating discrepancies and schedule planning difficulties.
- **High Vulnerability to Manual Errors**: Manual record-keeping relies entirely on human input, presenting a high operational risk of data mishandling. Staff face high administrative pressure during peak seasons, which frequently results in misspelled customer information, duplicate bookings for a single seat, or entirely misplaced physical files.
- **Inefficient Administrative Operations**: Compiling operational summaries, tracking daily cash flows, and managing seat distribution metrics requires substantial manual documentation. Without automated data centralization, retrieving historical booking histories or executing cross-counter itinerary adjustments becomes a time-consuming, unproductive chore for administrators.

3.0 Proposed Solutions

To eliminate the bottlenecks embedded in the manual workflow, the proposed web-based Bus Ticket Booking System introduces a computerized, centralized solution designed to automate end-to-end ticketing interactions. The primary interventions of the proposed solution include:

- **Remote Online Accessibility**: The system introduces an online platform compatible across various consumer electronics including smartphones, tablets, and computers. Customers can effortlessly book tickets anytime and anywhere, removing the obligation to wait in line at terminal counters.
- **Centralized Database & Real-Time Updates**: By utilizing a centralized database architecture, any ticket sale or schedule variation updates across the entire system instantly. Real-time seat tracking gives passengers clear transparency over exact layout allocations and completely eradicates the issue of double-booking.
- **Automated Data Validation**: To protect against human input errors, the platform embeds system-side entry validation forms. This ensures that passenger information remains clean and standardized while computing fare rates perfectly without manual calculation risks.
- **Streamlined Administrative Controls**: Bus operators gain access to an administrative dashboard configured for schedule generation, fleet tracking, and automated reporting. This centralized hub eliminates paper logs, reduces administrative workloads, and empowers data-driven management.

4.0 Current Business Process/Workflow

4.1 Overview of the current system:

The current system is entirely manual as it requires customers to physically visit the ticket counter to purchase bus ticket. At the same time, staffs are obligated to manage seat maps, printed schedules, and booking forms without a real-time tracking or digital payment options.

4.2 Stakeholder scenarios:

The following scenarios describe how the current system is used by key stakeholders:

Scenario 1: Customer books a ticket
A customer who wishes to travel from Kuala Lumpur to Johor Bahru must travel to the bus terminal. After queuing at the counter, the customer verbally provides travel details (route, date, time) to the staff. The staff checks the printed schedule and manually verifies seat availability on a paper chart. If seats are available, the staff fills out a paper booking form with the customer's details, collects cash payment, calculates change manually, and writes a handwritten ticket. The entire process can take 15-30 minutes, especially during peak periods.

Scenario 2: Counter Staff Manages Bookings
Counter staff must simultaneously handle walk-in customers, manage multiple paper booking forms, and update seat charts manually. During busy periods (school holidays, public holidays), staff face high pressure, leading to data entry errors such as incorrect passenger names, wrong seat numbers, or double-booking of the same seat. At end-of-day, staff must manually tally all bookings and prepare a summary report for the admin.

Scenario 3: Admin Reviews Booking Records
The admin must physically collect paper booking forms from each counter to compile daily reports. This process is time-consuming and error prone as some forms may be misplaced or illegible. There is no centralised view of overall seat occupancy, revenue, or booking trends. Any schedule changes must be manually communicated to all counter staff.

4.3 Workflow table (AS-IS)

The table below illustrates the step-by-step workflow of the current manual bus ticket booking process, including the actors involved and identified issues:

Step	Activity	Actor	Issues(AS-IS)
1	Customer travels to bus terminal ticket counter	Customer	Time-consuming

2	Customer queues to see counter staff	Customer	Long queues during peak hours and holidays
3	Staff asks for travel details	Counter Staff	Manual verbal communication; prone to errors
4	Staff manually checks physical schedule book	Counter Staff	No real-time updates; schedules may be outdated
5	Staff manually checks available seats on printed chart	Counter Staff	Risk of double booking due to simultaneous sales
6	Staff records passenger details on paper booking form	Counter Staff	Handwriting errors, risk of lost forms
7	Customer pays in cash; staff calculates change manually	Counter Staff	No digital payment; calculation errors possible
8	Staff manually writes out ticket and hands to customer	Counter Staff	No digital record; ticket can be lost or forged
9	Staff updates seat chart manually and files paper form	Counter Staff	No centralized database; admin retrieval difficult
10	End of day: staff manually tallies bookings and reports to admin	Counter Staff	Time-consuming; errors in daily reports common

5.0 Logical DFD (AS-IS)

This section presents the Logical Data Flow Diagram for the current AS-IS Bus Ticket Booking System. As established in Phase 2, the existing system is entirely manual, with customers required to visit the bus terminal counter to purchase tickets. The Context Diagram below provides the highest-level view of the system, treating the entire manual booking process as a single process interacting with the Customer, Staff, and Admin. Diagram 0 then expands this single process into its major activities, showing how travel details are collected, schedules and seats are checked, payment is recorded, tickets are issued, and a daily report is compiled, all without the support of computerised data storage.

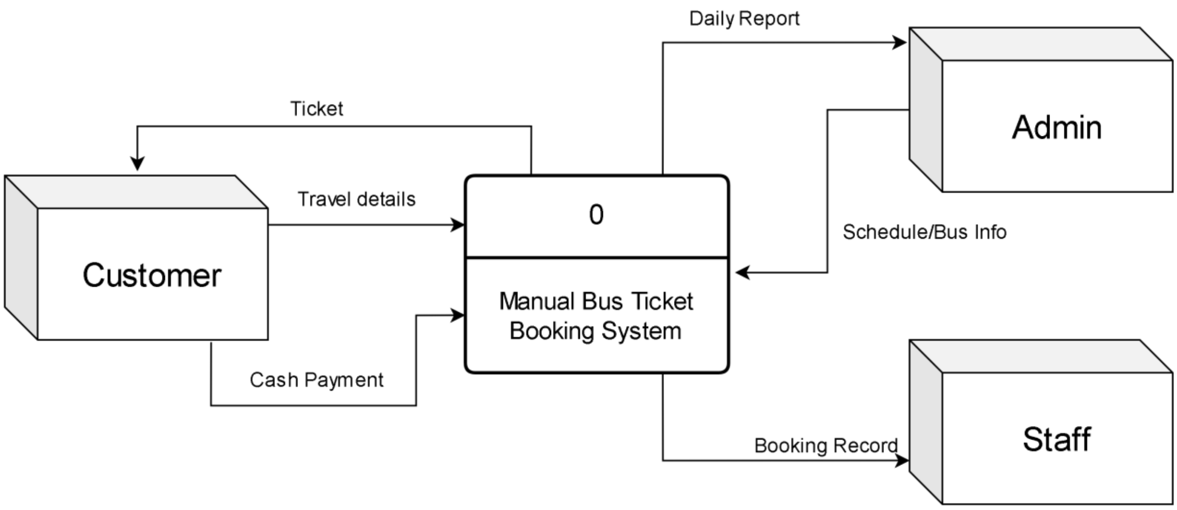
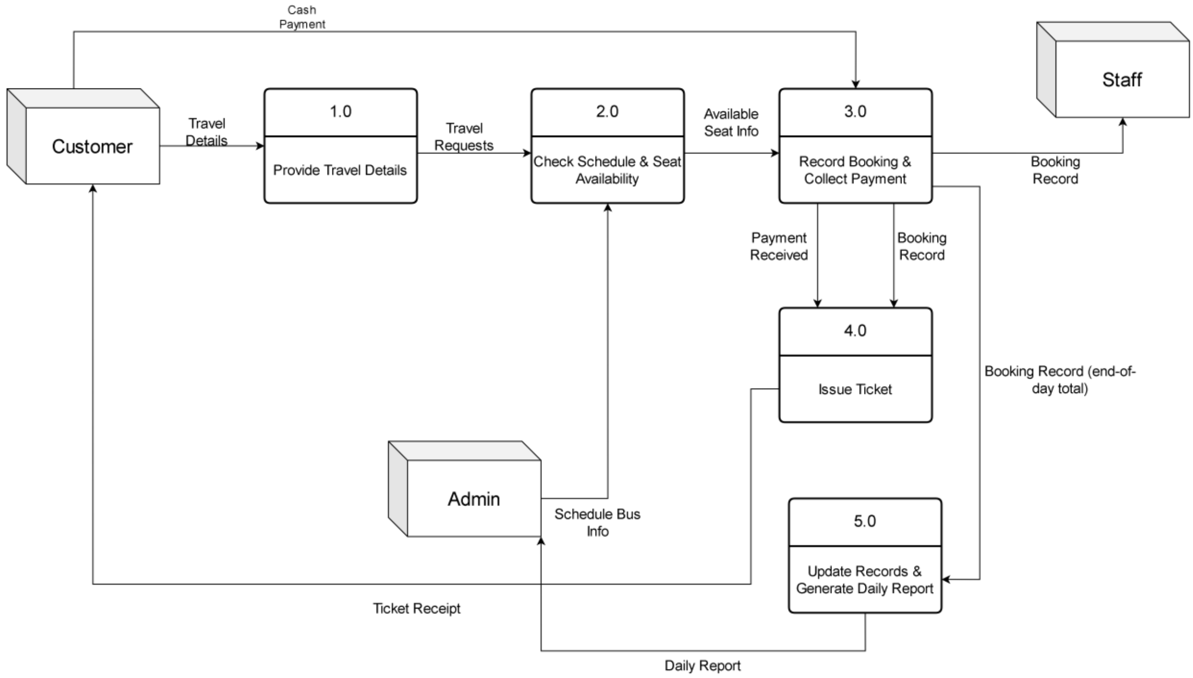


Diagram 0 AS-IS :



Based on the diagrams above, it is evident that the AS-IS system depends heavily on human intervention at every stage, from manually checking the printed schedule and seat chart to compiling the end-of-day report by hand. Each handoff between Customer, Staff, and Admin introduces a point where delays or errors can occur, which directly supports the problems identified earlier in the Problem Statement, namely long queues, double bookings, and inaccurate record-keeping. This AS-IS model therefore serves as the baseline against which the improvements of the proposed TO-BE system, discussed in the following section, can be measured.

6.0 System Analysis and Specification

6.1 Logical DFD TO-BE system (Context Diagram, Diagram 0, Child)

Context Diagram :

Having modelled the current manual process, this section presents the Logical DFD for the proposed TO-BE Online Bus Ticket Booking System. The TO-BE model was developed by refining the AS-IS process based on the requirements gathered through interviews and the questionnaire survey conducted in Phase 2, particularly the demand for real-time seat availability, online payment, and e-ticketing. The Context Diagram introduces a new external entity, the Payment Gateway, reflecting the system's move from manual cash handling to digital payment processing.

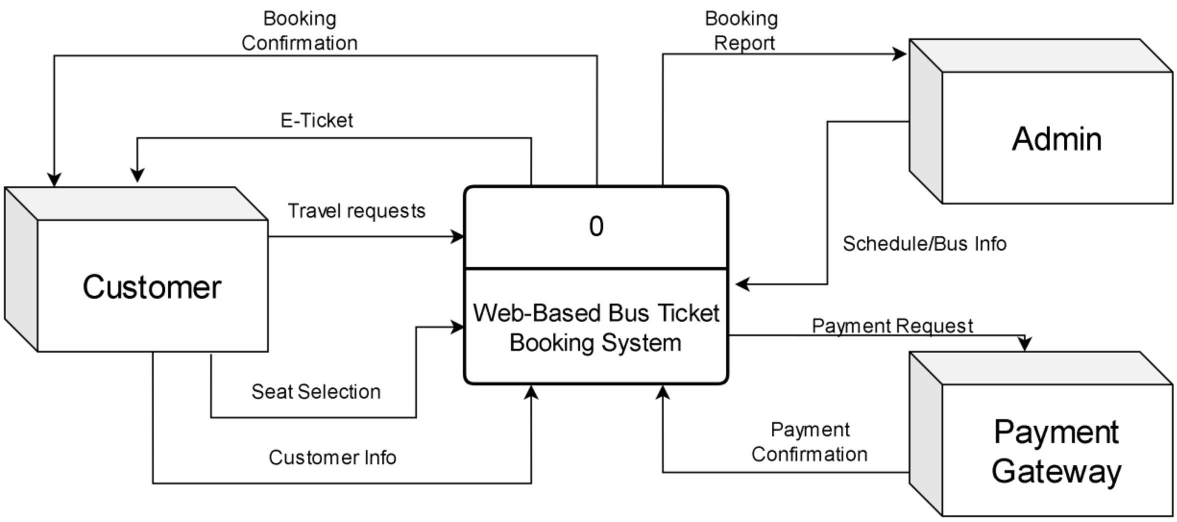
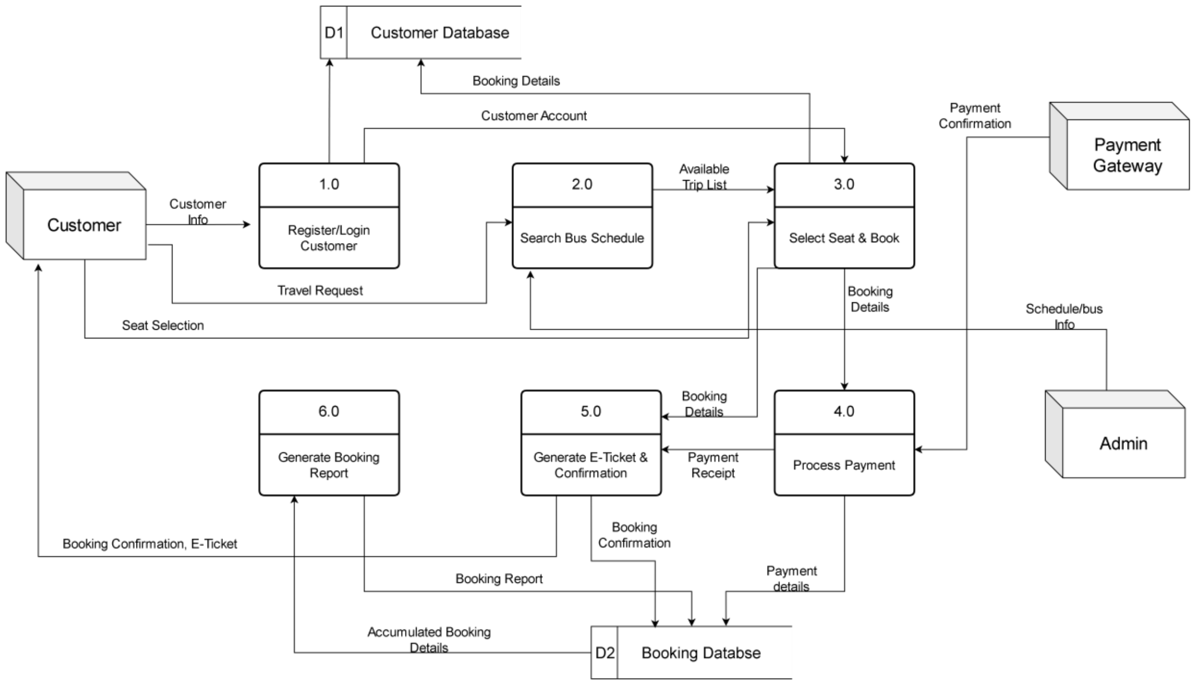


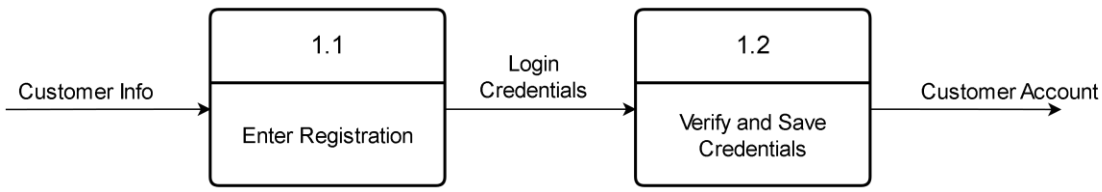
Diagram 0 :

Diagram 0 shown decomposes the proposed system into six major processes: Register/Login Customer, Search Bus Schedule, Select Seat & Book, Process Payment, Generate E-Ticket & Confirmation, and Generate Booking Report. Each of these processes is critical to the proposed system and is therefore further decomposed into a child diagram, presented below, to clarify the detailed steps and data store interactions involved.



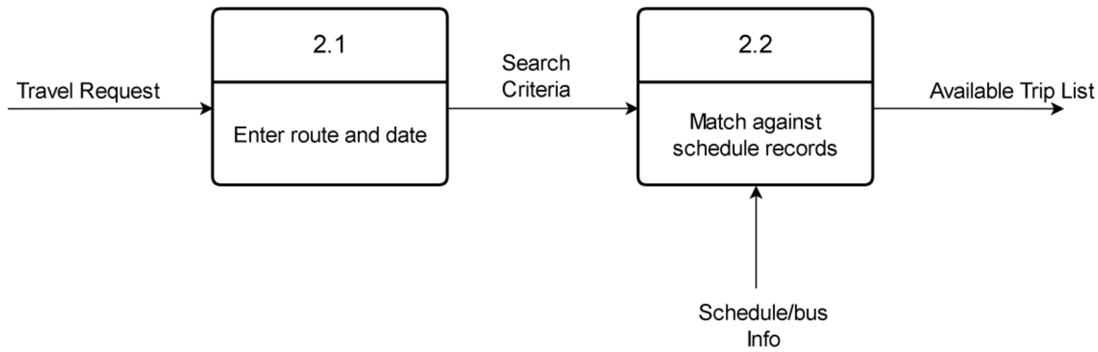
Compared to the AS-IS system, the TO-BE model replaces manual, paper-based activities with automated processes supported by the Customer Database (D1) and Booking Database (D2). This allows seat availability and booking status to be updated and retrieved in real time, removing the risk of double booking that was present in the manual system. The introduction of the Payment Gateway as an external entity also eliminates the need for manual cash handling and fare calculation, while the automatic generation of the booking report in Process 6.0 replaces the manual end-of-day tally previously performed by staff. Overall, the TO-BE DFD demonstrates how each weakness identified in the AS-IS analysis has been directly addressed through automation and centralised data management. **Child Diagram :**

i. Process 1.0: Register/Login Customer



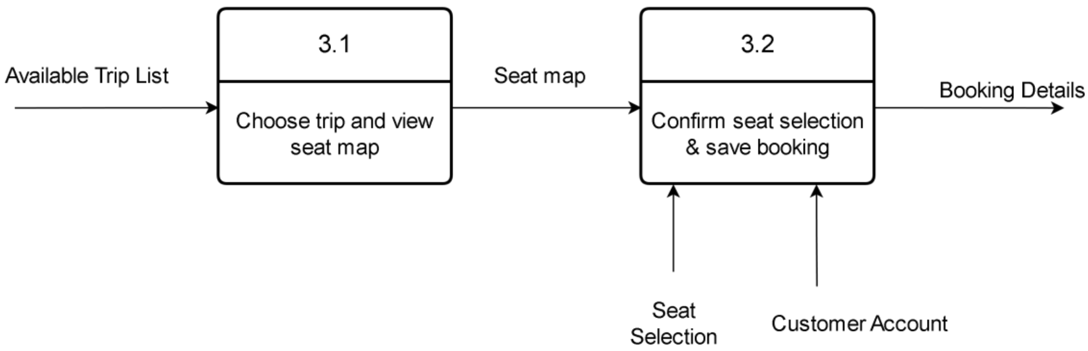
Process 1.0 – Register/Login Customer: This child diagram details how customer information is captured during registration and verified during login, with confirmed records stored in the Customer Database (D1)

ii. Process 2.0: Search Bus Schedule



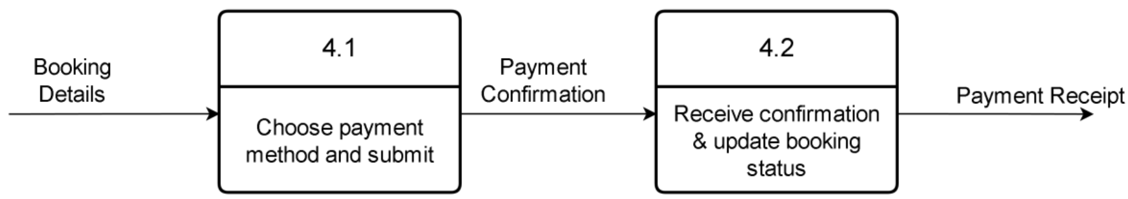
Process 2.0 – Search Bus Schedule: This child diagram shows how a customer's travel request is matched against the available trip records stored in the Booking Database (D2).

iii. Process 3.0: Select Seat & Book



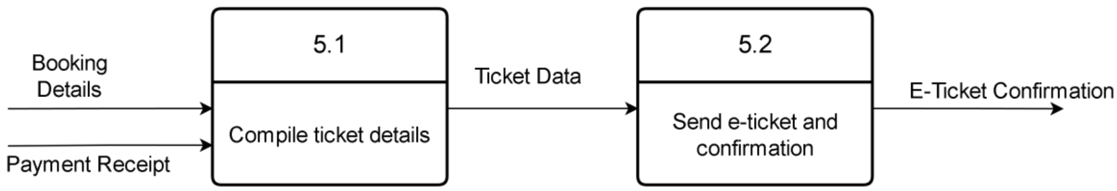
Process 3.0 – Select Seat & Book: This child diagram illustrates how a customer views the seat map for a chosen trip and confirms a booking, which is then recorded in the Booking Database.

iv. Process 4.0: Process Payment



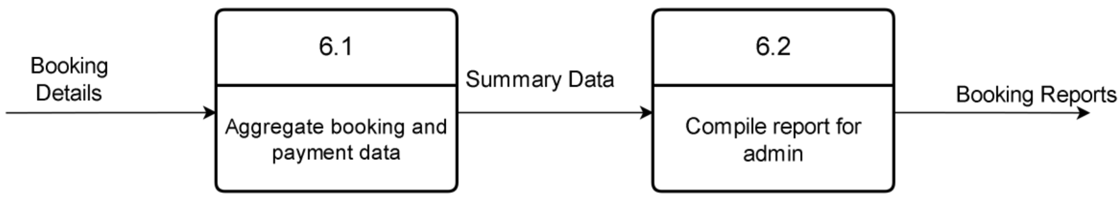
Process 4.0 – Process Payment: This child diagram explains how the booking is submitted for payment and how the booking status is updated once payment is confirmed by the Payment Gateway.

v. Process 5.0: Generate E-Ticket & Confirmation



Process 5.0 – Generate E-Ticket & Confirmation: This child diagram shows how confirmed booking and payment data are compiled into a ticket and sent to the customer as an e-ticket with confirmation.

vi. Process 6.0: Generate Booking Report



Process 6.0 – Generate Booking Report: This child diagram details how booking and payment data are aggregated over time and compiled into a report for the admin.

The DFD developed in this phase differs from the one produced in Phase 2 because Phase 3 is built on a more complete understanding of the system, gathered through the requirement analysis, functional and non-functional requirements, and stakeholder scenarios documented earlier in this report. While Phase 2 focused on identifying the AS-IS process and a general idea of what the proposed system should achieve, Phase 3 incorporates that additional analysis into a refined, logical TO-BE model. As more information became available regarding the required inputs, processes, and outputs of the system, certain processes were consolidated, new external entities such as the Payment Gateway were introduced, and data stores were defined to reflect how data would actually be managed and retrieved. This evolution from a general manual workflow to a detailed, structured logical model demonstrates how the system analysis process matures as more information is gathered and validated with the requirements covered throughout the syllabus.

6.2 Process Specification (based on Logical DFD TO-BE)

For this project we decide to do in English Structure process specification

1. Customer Registration and Login

```
DO WHILE Customer attempts to Register
        Read customer Email, Password
        IF Email already exists
                Display "Email Already Registered"
        ELSE
                INSERT new data at D1
        ENDIF
ENDDO
```

2. Travel Request

```
DO WHILE Customer interacts with Booking Dashboard
Read Departure Station, Destination Station, Travel Date
Read data from admin
        IF matching schedules are found THEN
                List all available trips
                Display trip information
        ELSE
                Display "No Trips Available for Selected Criteria"
        ENDIF
ENDDO
```

3. <u>Process Ticket Booking</u>

```
DO WHILE Customer selects a specific trip

Generate dynamic interactive seat layout map

    Read reserved seat flags

    Display available seats as clickable indices, available (vacant), not available
    (busy)

    IF Customer selects an available Seat Number THEN

        Temporarily lock Seat Number

        Verify all required passenger fields are non-empty

        Calculate Trip Fare

            Total Ticket Price = Base Fare + Service Tax
        Generate E-tickets based on trip information

        FORWARD Booking ID and Total Ticket Price payload to Process 4.0
    ELSE

        Display "No Seat Selected"

    ENDIF

ENDDO
```

4. Payment Process
   DO WHILE Booking Payload is active and 10-minute timer has not expired

        Display Payment Method Options (FPX Online Banking, Credit Card, E-Wallet)

        Customer selects Payment Method

        REDIRECT secure transaction token payload to External Payment Gateway API

   IF External Payment Gateway returns "SUCCESS" Callback Token THEN

        UPDATE status attribute

        TRIGGER Process 5.0 (Generate Ticket)

   ELSE

        UPDATE status attribute to "Cancelled/Failed"

            Display "Payment Failed. Please try again." Message

   ENDIF

   ENDDO

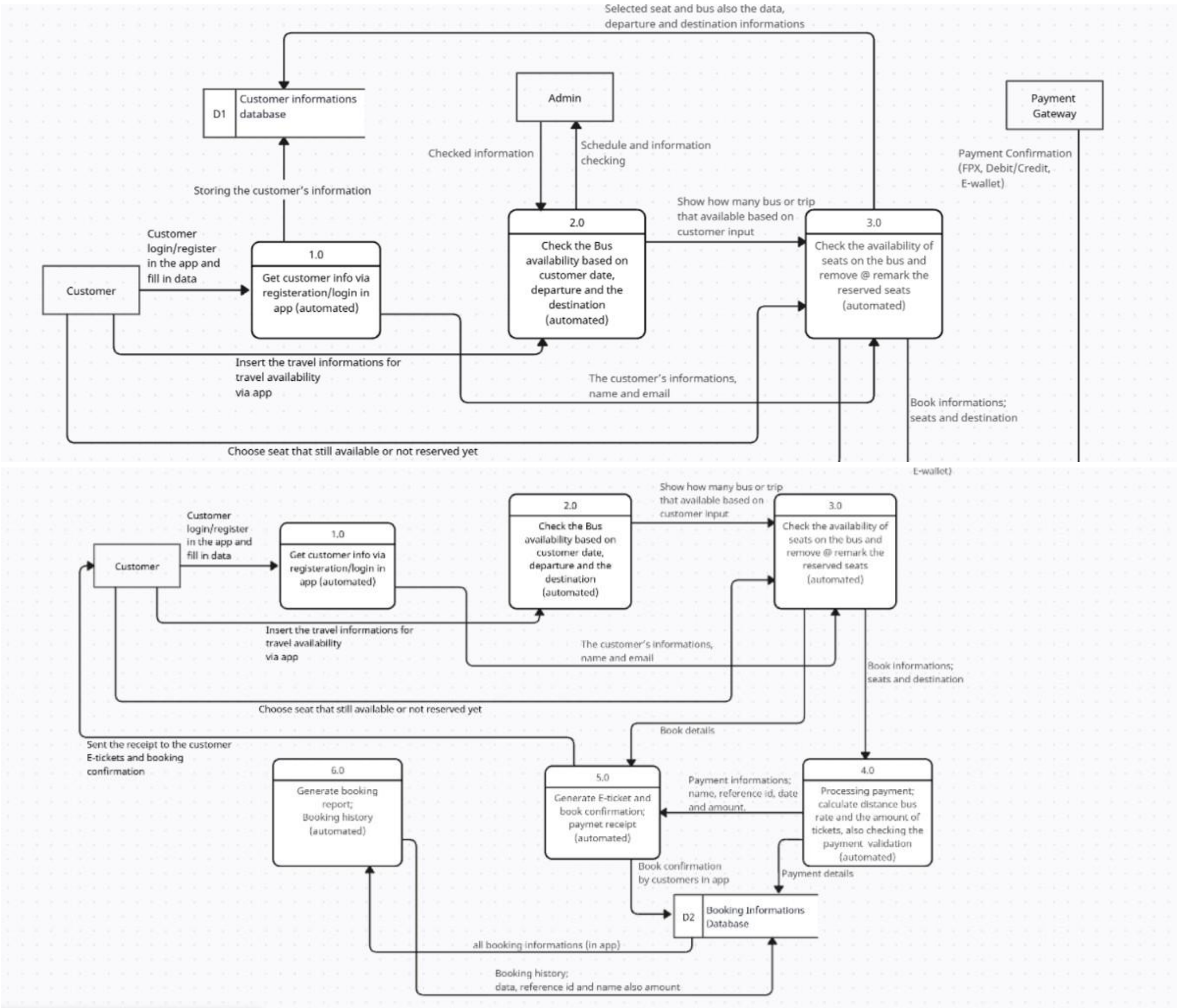
5. Generate E-tickets and Confirmation
   DO WHEN Triggered by successful payment verification
     Read complete details
     Compile alphanumeric structural core strings into a unique string
     Generate graphical QR Code containing encrypted Booking validation token
   ENDDO

6. Booking Reports @ History
   DO
        Read booking details, payment details
        Generate history of booking
        Generate booking report
        UPDATE to admin @ company file
   ENDDO

7.0 Physical System Design

7.1 Physical DFD TO-BE system

1. 0 Diagram

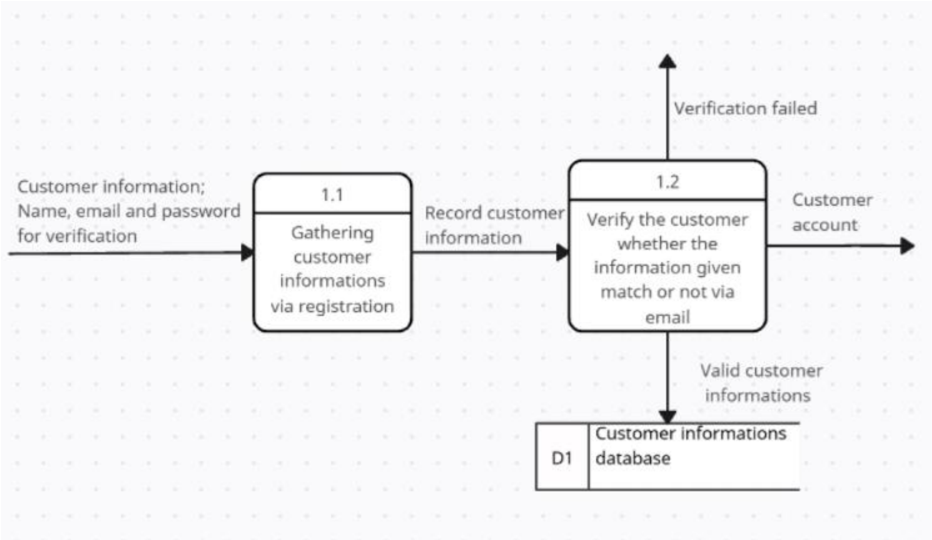


*this error due to limitations of elements usage in the free website, so I must several elements.

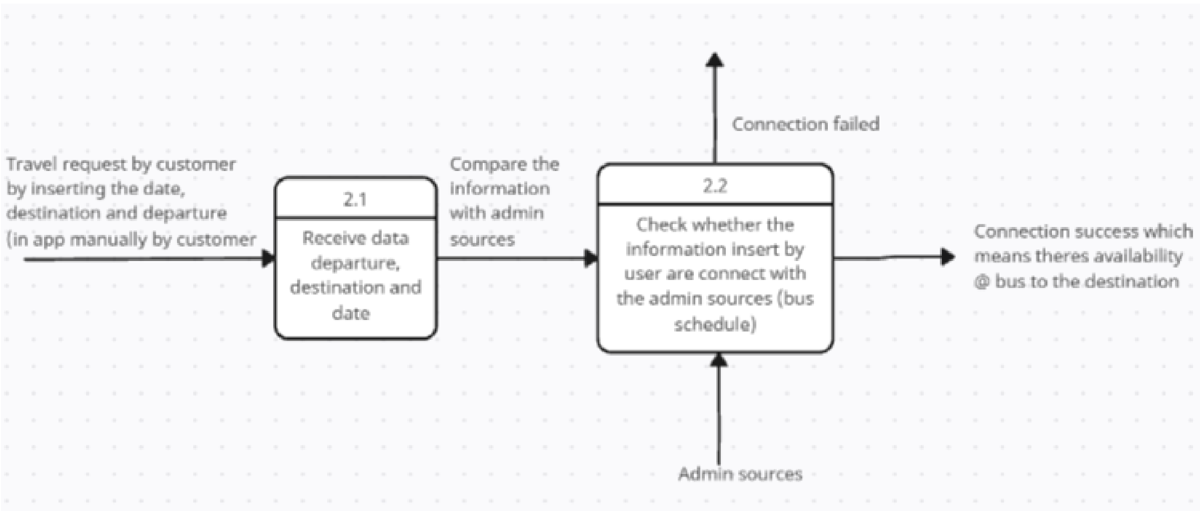
Payment gateway -> 4.0 process

2. Child Diagram

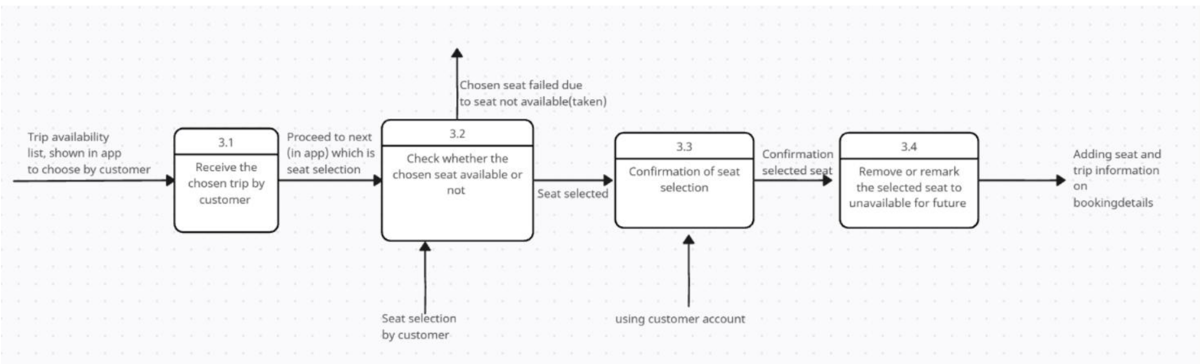
Process 1:



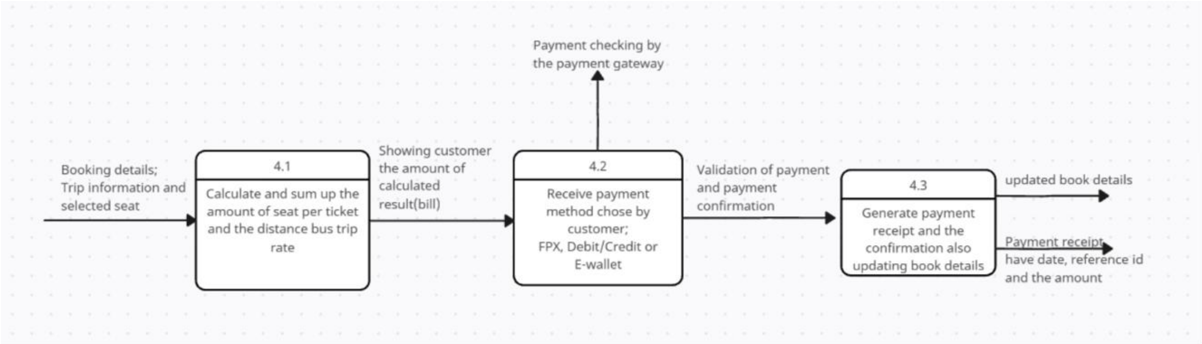
Process 2:



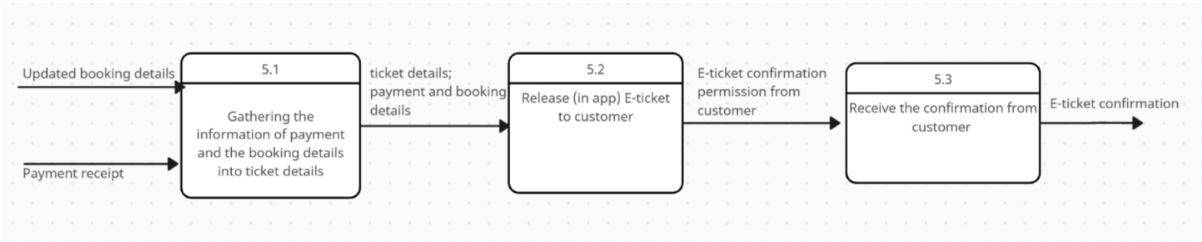
Process 3:



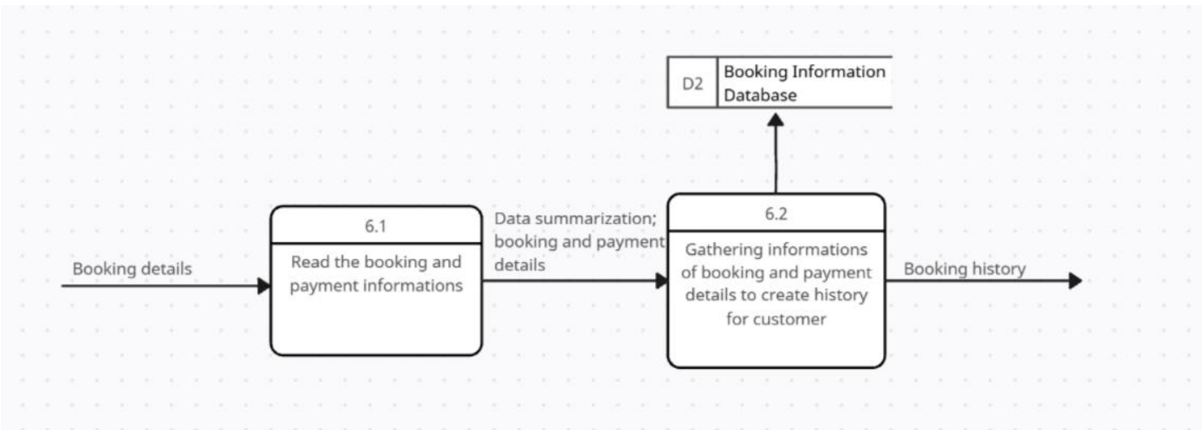
Process 4:



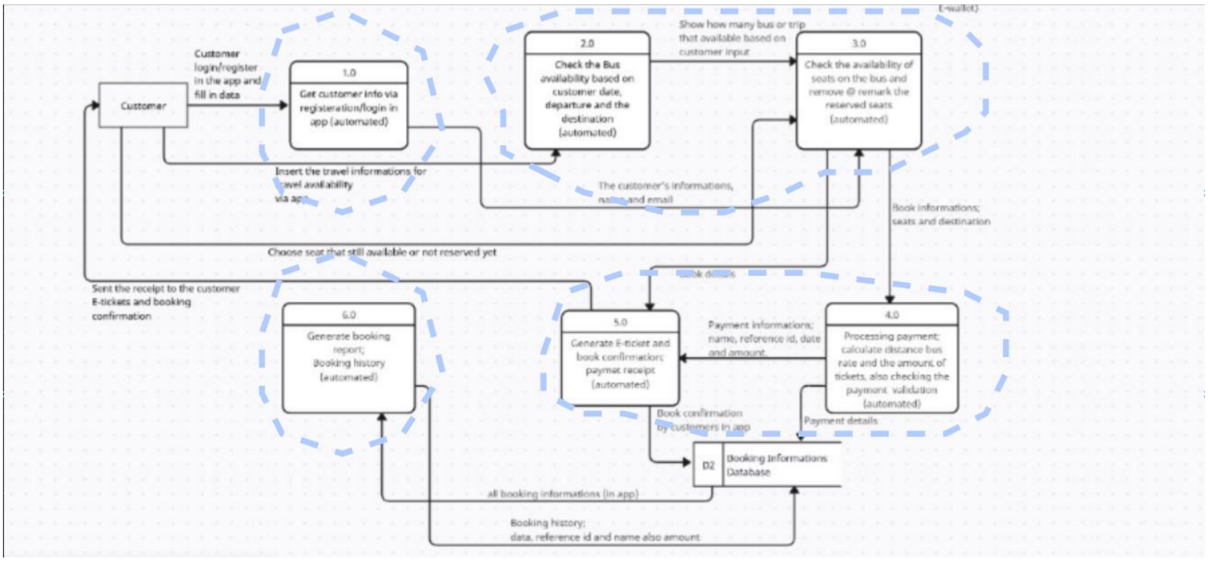
Process 5:



Process 6:



3. Partitioning



Partitioning is the process of examining a data flow diagram and determining how it should be divided into collections of manual procedures and computer programs. A dashed line is drawn around a process or group of processes that should be placed in a single computer program. However, some programs are classified as batch programs as this processes have their own input and output as well as its own timing. Meanwhile, partition are done on Process 2.0 and Process 3.0 for instance as they have similar tasks and consistency of data that are transferred.

4. CRUD Matrix

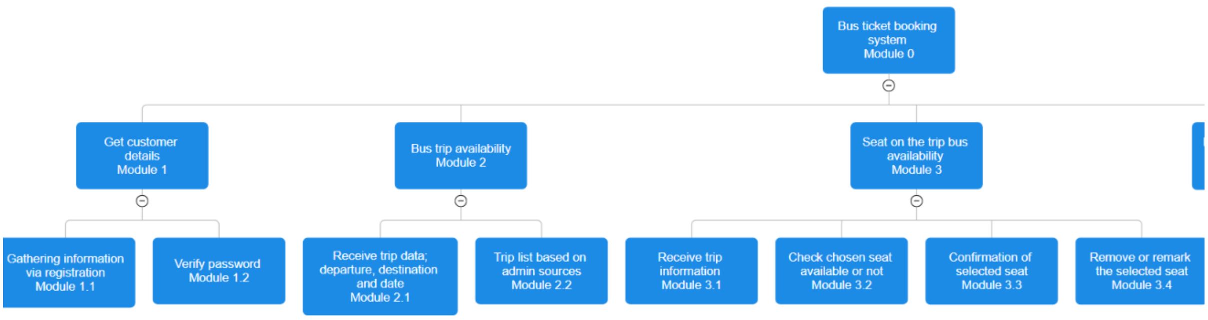
C-Create, R-Read, U-Update, D-Delete

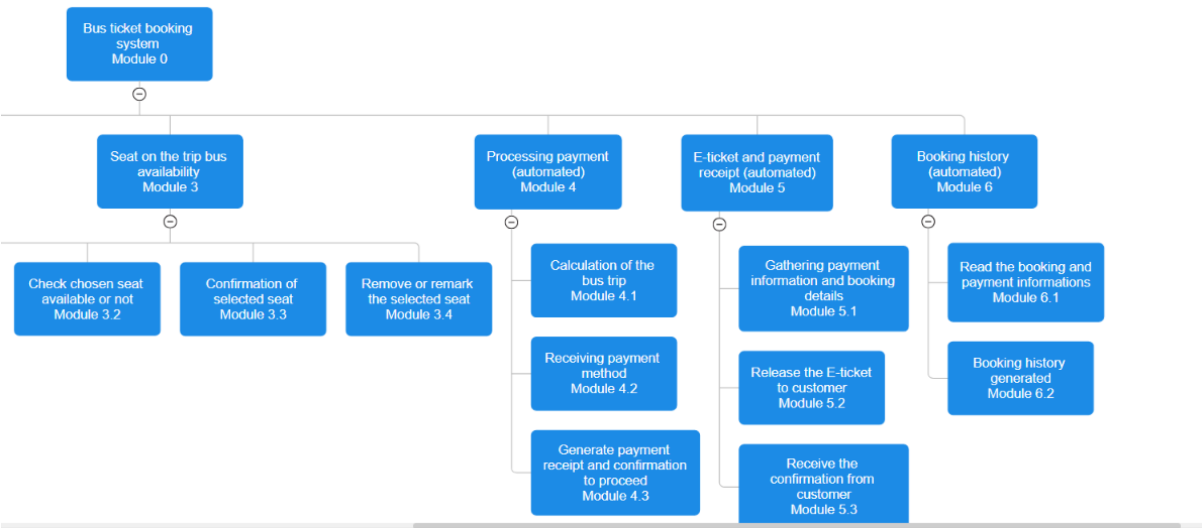
Activity	Customer	Booking
Customer registration and login	R	
Trip requesting	C	
Select trip	U, D	
Select seat	C, U, D	
Confirmation		R, U
Make payment		U
Generate E-tickets and receipt		C, U
Payment confirmation		R
Generate payment history		C, U

5. Event Response Table

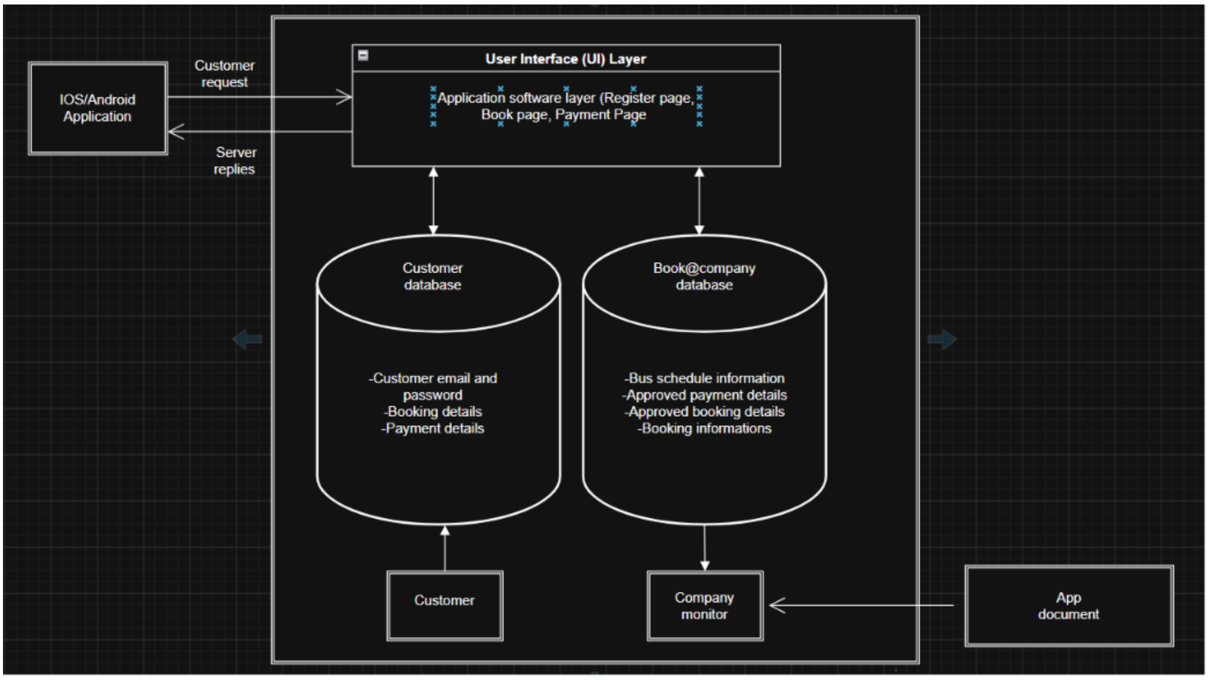
Event	Source	Trigger	Activity	Response	Destination
Customer register and login	Customer	Customer name, email, password	Verify password and the email	If success, welcome page (in app)	Customer
Customer request trip	Customer	Departure, Destination and Date	Check the availability	Trip lists	Customer
Customer select seat	Customer	Click on seat icon	Check the availability seat	Seat option page	Customer
Customer confirmation trip	Customer	Click “Proceed to payment”	Trip and seat details printed	Confirmation page	Customer and Admin
Obtain customer payment	Customer	Customer payment method	Verify the payment at the payment gateway	Payment gateway page	Customer and payment gateway company

6. Structure Chart





7. System Architecture



Customer
request
IOS/Android
Application
Server
replies
User Interface (UI) Layer
Application software layer (Register page,
Book page, Payment Page
Customer
database
-Customer email and
password
-Booking details
-Payment details
Book@company
database
-Bus schedule information
-Approved payment details
-Approved booking details
-Booking informations
Customer
Company
monitor
App
document

1. User Interface (UI) Layer

   Software Application:

   - Platforms: IOS and Android

   Feature:

   - **Core Software Flow:** 1. The customer interacts with features like User Registration, User Login, or the Welcome/Booking Dashboard.
   - The application bundles these actions into secure network requests (**Customer Requests**) and transmits them externally via standard web protocols.
   - Once processing is complete, the interface receives backend payloads (Server Replies) to dynamically render tickets, update seat maps, or display login confirmations.
2. Application Layer
   - It listens for incoming API communication traffic from the iOS/Android client applications.
   - It parses the client requests, applies security checks, and generates corresponding query sequences to read from or write to the storage layer.
3. Data storage Layer
   - It handles transaction logs, user account profiles, bus schedules, and invoice records.
   - It executes incoming transactional queries from the application server and returns the requested data matrices safely back to the server tier.

8.0 System Wireframe (Input Design, Output Design)

8.1 Introduction

For our project prototype, we used Figma as the main software to showcase layout and flow of our app. The interface is designed to be mobile-first to accommodate users on the go. The system splits into two major components which are the Customer Interface and the Admin Interface. In detail, the Customer Interface is used for ticket booking and digital payment processes, while the Admin Interface is for fleet management and report generation.

8.2 Input Design Documentation

Input design focuses on data entry and how users input information into the system efficiently while minimizing errors.

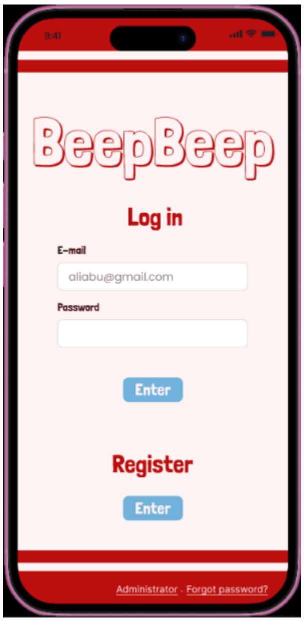


Figure 1: User Log in Page

UI Component	Input Type	Purpose	Validation
Email field	Text field	Captures the user's registered email	Must match a valid email; cannot be left blank
Password field	Text field	Captures the secure password associated with the account	Requires a minimum of 8 characters
Log in "Enter" button	Command button	Submits the E-mail and Password input to authenticate the user.	Disabled if input fields are empty; triggers an error message if the account credentials do not match database records.

Register "Enter" button	Command button	Redirects new users to the sign-up/registration form screen.	None; immediately clears current inputs and shifts the viewport to the Registration page.
Administrator Link	Hyperlink	Redirects admins to the dedicated back-end administration portal.	Routes users to the admin validation interface.
Forgot Password? Link	Hyperlink	Directs users to a recovery page if they cannot remember their credentials.	Triggers a secure verification/OTP layout to the user's email.

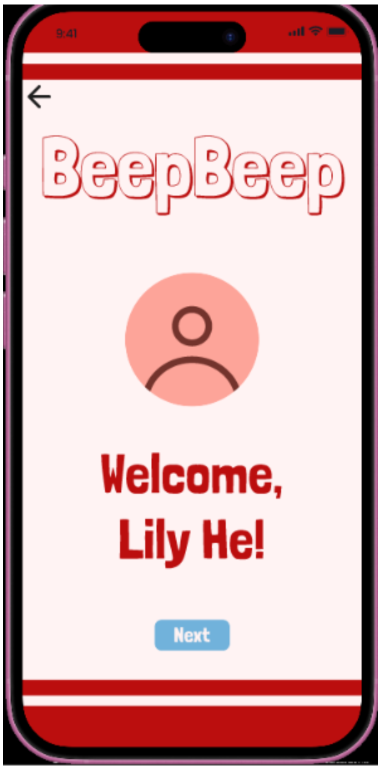


Figure 2: Successfully log in layout

UI Component	Input Type	Purpose	Validation
"Next" button	Command button	Directs user to home page	None
Go back arrow	Command button	Redirects user back to the previous page which is the log in page	None

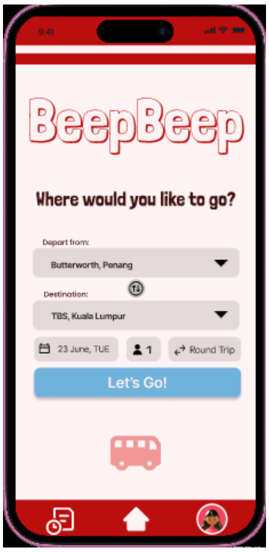


Figure 3: Home page

UI Component	Input Type	Purpose	Validation
Depart from:	Dropdown Selector	Captures the user’s desired terminal departure location	Restricts input to valid terminal locations served by the company network to eliminate typing errors
Swap Route Icon	Tonal Toggle Button	Instantly interchanges the values currently selected in the “Depart from” and “Destination” fields	Operational convenience feature; speeds up user workflow for reverse routes
Destination:	Dropdown Selector	Captures the user’s final arrival station	Cannot be identical to the “Depart from” selection; dynamically lists available connecting hubs
Date Selection Box	Calendar Picker Modal	Captures the intended travel date	Restricts past dates from being selected
Passenger Count	Numeric Spinner	Captures the total number of tickets required for the trip	Defaults to 1 passenger
Trip Type Toggle	Segmented Control Button	Allows the user to select between a "One-Way" or a "Round Trip" journey structure	Selecting "Round Trip" dynamically generates a secondary return date calendar picker field

"Let's Go!" Button	Command Action Button	Submits the consolidated search parameters to look up corresponding bus schedule results	Disabled if location fields are left unselected; executes a database search query upon activation
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home (current view), and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

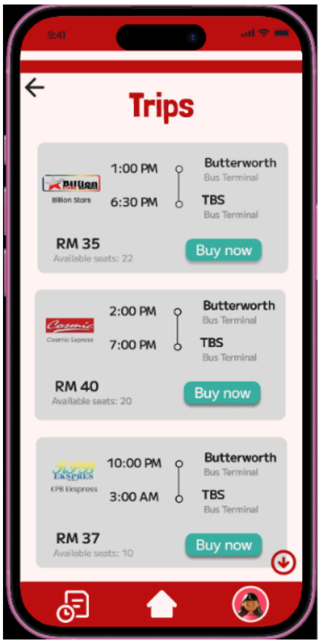


Figure 4: Search trip page

UI Component	Input Type	Purpose	Validation
“Buy Now” Button	Command Action Button	Captures the user’s desired bus trip.	Restricts the following trip details to those only of the desired bus trip.
Go back arrow	Command Button	Redirects user to the previous page, which is the Home page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home (current view), and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

Figure 5: Seat selection page (unselected seat) and error dialog boxes

UI Component	**Input Type**	**Purpose**	**Validation**
Seats	Tonal toggle Button	Allows the user to select and deselect their desired seats.	Disabled if the seats are occupied or currently selected by someone else.
“Your Seats” Button	Command Action Button	Opens an overlay that displays the number of the user’s selected seats, total ticket price (including tax) and number of tickets selected.	Disabled if no seats are selected. Interacting with the disabled button opens a dialog box to inform the user to select a seat first.
“Next →” Button	Command Button	Submits the selected seats, then directs user to Trip Confirmation page to continue the payment process.	Disabled if no seats are selected. Interacting with the disabled button opens a dialog box to inform the user to select a seat first.
“X” Button	Command Button	Closes dialog box	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Trips page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

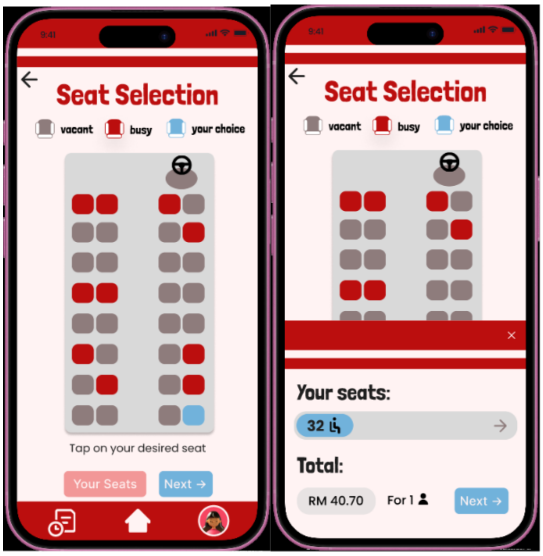


Figure 6 and Figure 7: Seat selection page (selected seat) and Your Seats overlay

UI Component	Input Type	Purpose	Validation
Seats	Toggle Button	Captures the user's desired seats.	Disabled if the seats are occupied or currently selected by someone else.
"Your Seats" Button	Command Action Button	Opens an overlay that displays the number of the user's selected seats, total ticket price (including tax) and number of tickets selected.	Disabled if no seats are selected. Interacting with the disabled button opens a dialog box to inform the user to select a seat first.
"Next →" Button	Command Button	Submits the selected seats, then directs user to Trip Confirmation page to continue the payment process.	Disabled if no seats are selected. Interacting with the disabled button opens a dialog box to inform the user to select a seat first.
"X" Button	Command Button	Closes overlay	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Trips page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

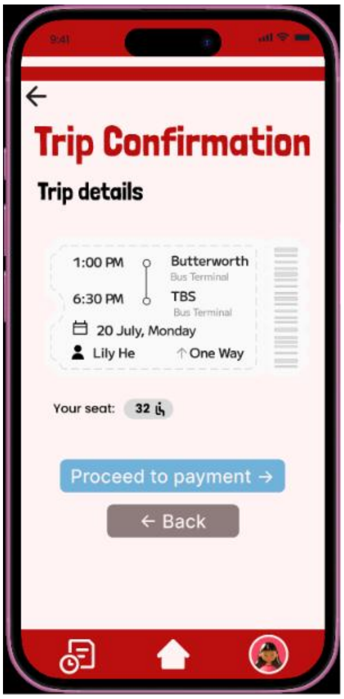


Figure 8: Trip Confirmation Page

UI Component	**Input Type**	**Purpose**	**Validation**
“Proceed to payment →” Button	Command Button	Confirms the user’s selection, then directs the user to the Payment page.	None
“← Back” Button	Command Button	Redirects user to the previous page, which is the Seat Selection page.	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Seat Selection page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

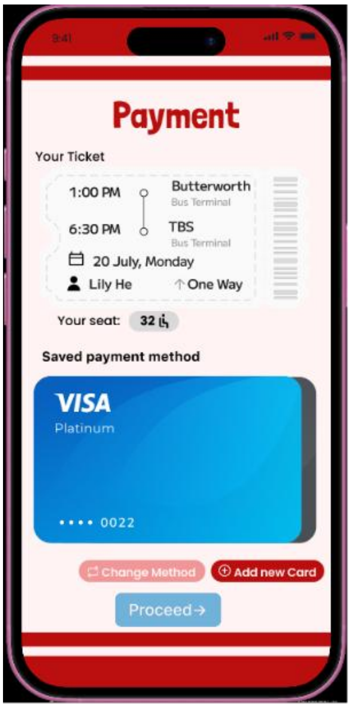


Figure 9: Payment page

UI Component	Input Type	Purpose	Validation
Card Button	Tonal Toggle Button	Allows the user to cycle through saved card payment methods.	Disabled if there is only one saved card payment method.
"Proceed →" Button	Command Button	Confirms the user's payment method, then directs the user to the next Payment page.	Disabled if no payment method is selected.
"Change Method" Button	Command Button	Opens an overlay that allows the user to change their payment method.	None
"Add new Card" Button	Command Button	Enables the user to add an additional card payment method.	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Seat Selection page.	None

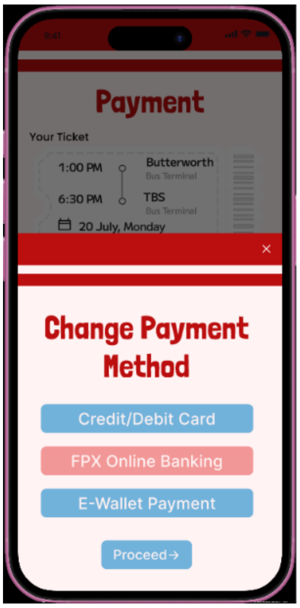


Figure 10: Change Payment Method overlay

UI Component	Input Type	Purpose	Validation
"Credit/Debit Card" Button	Command Action Button	Change user's payment method to a credit or debit card, then prompts user to enter their card's details.	None
"FPX Online Banking" Button	Command Action Button	Change user's payment method to FPX online banking method, then redirects user to their selected bank's FPX page.	None
"E-Wallet Payment" Button	Command Action Button	Change user's payment method to an e-wallet of their choice, then prompts user to enter their e-wallet details.	None
"X" Button	Command Button	Closes overlay	None
"Proceed →" Button	Command Button	Confirms the user's selection, then directs the user to the next Payment page.	None

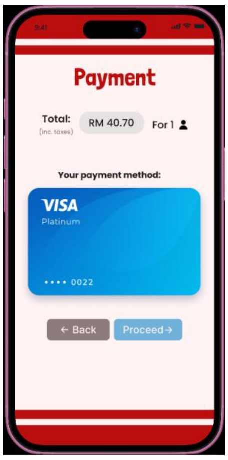


Figure 11: Payment page

UI Component	Input Type	Purpose	Validation
“Proceed →” Button	Command Action Button	Confirms the user’s selection, then opens an overlay that prompts the user to verify their identity.	None
“← Back” Button	Command Button	Redirects user to the previous page, which is the Payment page.	None

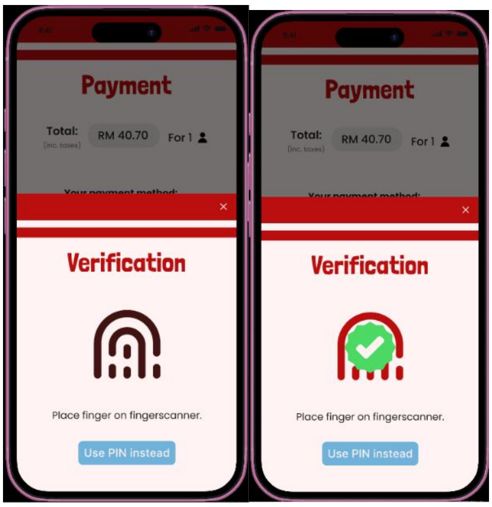


Figure 12 and Figure 13: Verification using biometrics, before and after verification.

UI Component	Input Type	Purpose	Validation
"Use PIN instead” Button	Command Button	Change verification method to PIN number.	Disabled if the user has not set up a PIN number to use as verification.
“X” Button	Command Button	Closes overlay	None

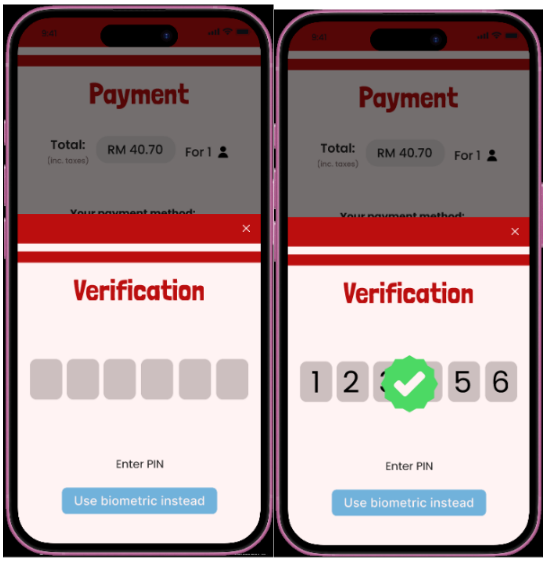


Figure 14 and Figure 15: Verification using PIN number, before and after verification.

UI Component	**Input Type**	**Purpose**	**Validation**
PIN number field	Numerical field	Accepts a 6-digit PIN number set by the user; upon completing entering the PIN number, submits the input to complete processing the payment.	No action occurs when incorrect PIN number is entered; user must re-enter the correct PIN number to complete processing the payment.
"Use biometric instead” Button	Command Button	Change verification method to biometric.	Disabled if the user has not set up biometric to use as verification.
“X” Button	Command Button	Closes overlay.	None

Figure 16: Payment Successful page

UI Component	**Input Type**	**Purpose**	**Validation**
"Go to Booking History →" Button	Command Button	Directs user to Booking History page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

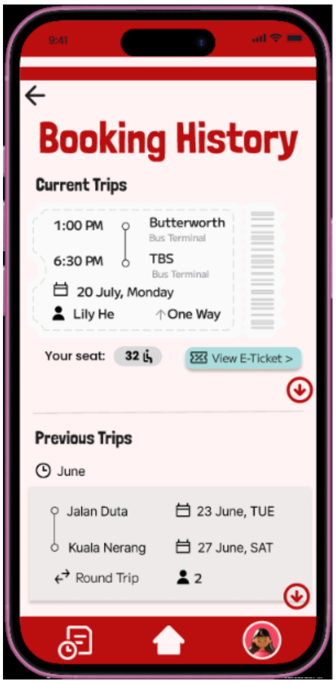


Figure 17: Booking History Page

UI Component	Input Type	Purpose	Validation
"View E-Ticket" Button	Command Button	Directs user to E-Ticket page.	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Home page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings (current view), Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

Figure 18: E-Ticket page

UI Component	Input Type	Purpose	Validation
"Cancel Booking" Button	Command Button	Directs user to Cancel Booking page.	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Home page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings (current view), Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

Figure 19: Cancel Booking

UI Component	Input Type	Purpose	Validation
“Yes” button	Command button	Confirms cancel booking and directs user to booking history page	None
“No” button	Command button	Cancels the Cancel Booking action and directs user back to booking history page.	None

Figure 20 and Figure 21: User Profile page

UI Component	Input Type	Purpose	Validation
Edit Button	Command Button	Allows user to change and submit changes to personal information.	None
Full Name, Identification No., E-mail and Phone Number Field	Text Field	Captures the user's related personal information	None
Birthdate	Calendar Picker Modal	Captures the user's birth date.	Restricts future dates and dates more recent than 10 years ago.
"Change Password" Button	Command Button	Directs user to change password page	None
"Payment Verification Method" Button	Command Button	Directs user to Payment Verification Methods page that allow user to update or remove biometrics and PIN number.	None
"Register as Administrator" Button	Command Button	Directs user to Register Administrator page	None
"Log out" Button	Command Button	Directs user to the Log Out page	None

Notification Bell Icon	Header Icon Button	Displays a real-time count of unread system updates to keep the user instantly informed	The numeric badge automatically resets to 0 once the notification feed overlay is opened by the user.
Customer Service Icon	Header Icon Button	Directs user to Customer Service page	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Home page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile (current view) settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

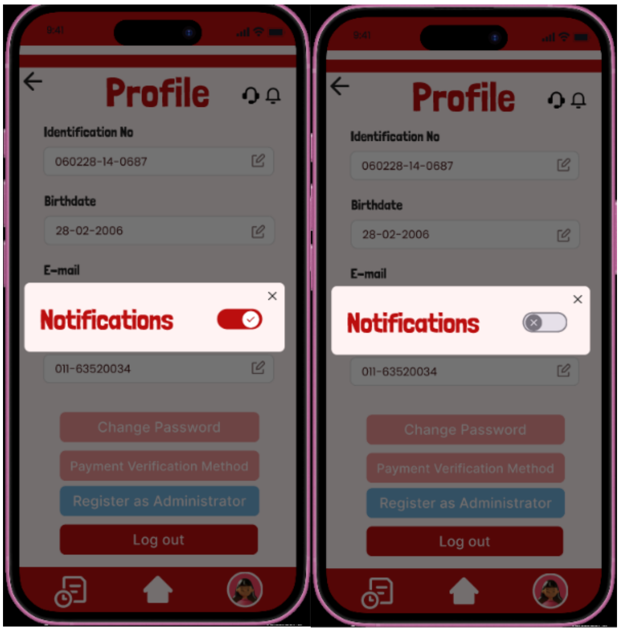


Figure 22 and Figure 23: Notification setting overlay

UI Component	**Input Type**	**Purpose**	**Validation**
“Notifications” toggle button	Modal Switch Toggle	Captures the user’s preference to allow or mute real-time mobile push notifications.	Binary Boolean constraint state
Modal Close Icon (X)	Dismiss button	Closes the notification overlay box to bring the user back to the primary profile layout	Instantly clears the overlay dimming layer without modifying

			background profile data variables.

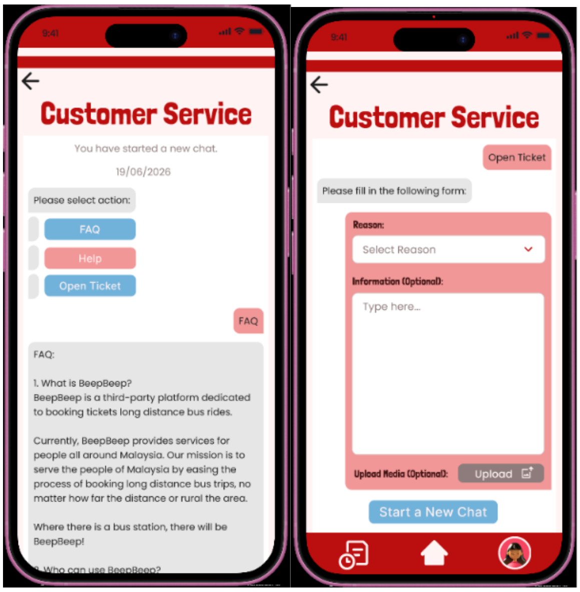


Figure 24 and Figure 25: Customer Service chat

UI Component	**Input Type**	**Purpose**	**Validation**
"FAQ" Button	Command Button	Triggers and displays the Frequently Asked Questions list within the chat interface	None
"Help" Button	Command Button	Opens the user manual or quick-start guide for assistance within the chat interface	None
"Open Ticket" Button	Command Button	Redirects the user to the ticket submission form screen	None
Reason field	Dropdown menu	Allows user to select the specific reason or category for opening a support ticket	Restricts user to pre-selected options
"Upload" Button	Command button / File Picker	Allows the user to attach an optional screenshot or file supporting their issue	Restrict file size and format
"Start a New Chat" Button	Command Button	Resets the current support session and	None

		redirects the user back to the main chat selection menu	
Information field	Text field	Allows the user to type in a detailed, free-text description of their issue or question	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Home page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings (current view), Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

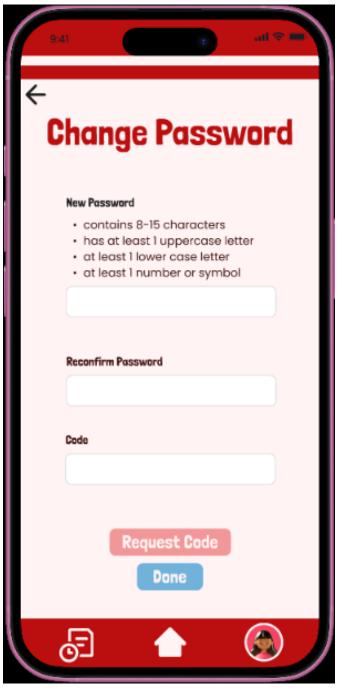


Figure 26: Change password while logged in page

UI Component	Input Type	Purpose	Validation
New Password	Text Box	Captures the new alphanumeric security password intended for account modification	Must satisfy all listed regex criteria

Reconfirm Password	Text Box	Captures the new password entry a second time to ensure typing consistency	Must match the input provided in the New Password field exactly
Code	Text Box	Captures the security verification/OTP code sent to the user's registered communication channel.	Cannot be left blank; restricts input to the exact character length expected of the security token.
Request Code Button	Command Buton	Triggers the system backend to generate and dispatch a time-sensitive verification code.	Triggers a cooldown timer constraint
Done Button	Command Button	Submits the validated password strings and security code to commit changes to the backend database.	Disabled until all criteria are met

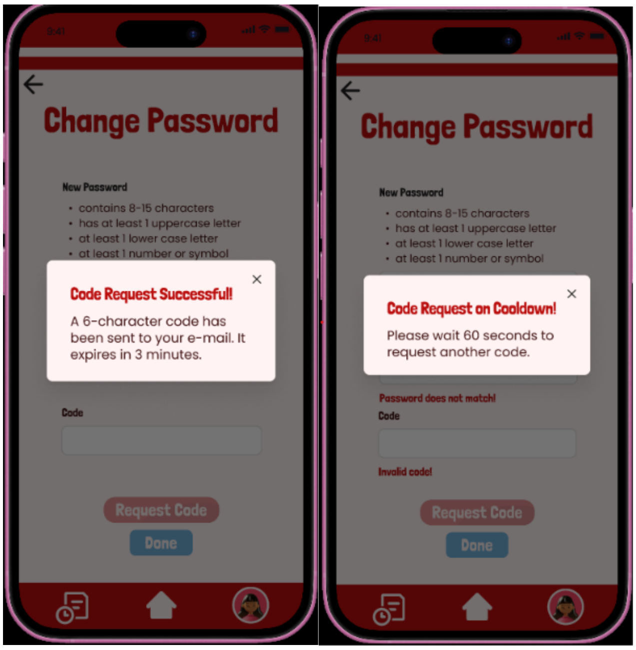


Figure 27: Request Code dialog boxes

UI Component	**Input Type**	**Purpose**	**Validation**
"X" Button	Command Button	Closes dialog box.	None

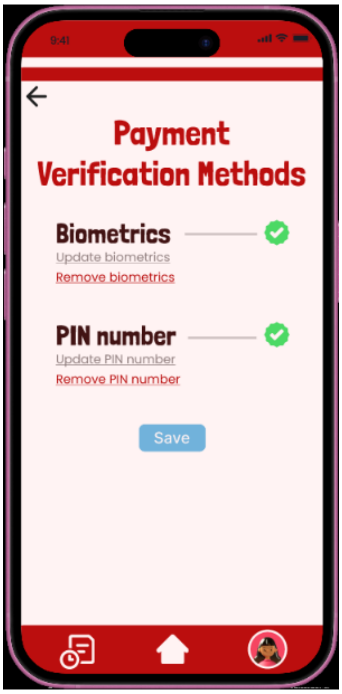


Figure 28: Payment Verification Methods page

UI Component	Input Type	Purpose	Validation
Update biometrics	Hyperlink	Opens the mobile OS-native biometric configuration interface to register or update biometric tokens.	Must successfully complete hardware-level verification via the phone's native security engine.
Remove biometrics	Hyperlink	Removes the registered biometric profile from being used as a transaction verification method.	Triggers a confirmation modal to verify the user wants to revert to basic password or PIN verification.
Update PIN number	Hyperlink	Redirects the user to a secure entry keypad screen to modify or create a new numeric PIN code.	Enforces specific numeric constraints
Remove PIN number	Hyperlink	Deactivates security PIN verification for rapid transaction flows.	Restricted constraint: The system will block removal if biometrics are also inactive, ensuring at least one secure verification method is preserved.

“Save” Button	Command Button	Commits all updated security verification preferences	None
Go back arrow	Command Button	Redirects user to the previous page, which is the Home page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings (current view), Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

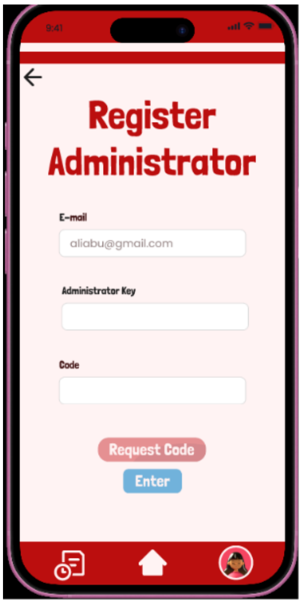


Figure 29: Register Administrator page

UI Component	**Input Type**	**Purpose**	**Validation**
Email field	Text field	Captures the user’s registered email	Must match a valid email; cannot be left blank
Administrator Key field	Text field	Captures a secure administrator key	Requires a minimum of 8 characters
Code field	Text field	Captures the temporary secondary verification OTP sent to the user’s secure communication channel.	Requires a minimum of 6 characters
“Enter” button	Command button	Submits	None

Go back button	Command button	Redirects admins to the dedicated back-end administration portal.	Routes users to the admin validation interface.
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

Figure 30: Log out confirmation page

UI Component	Input Type	Purpose	Validation
"Yes" button	Command button	Confirms log out and directs user to post-logout page	None
"No" button	Command button	Cancels the log out action and directs user back to user profile page.	None

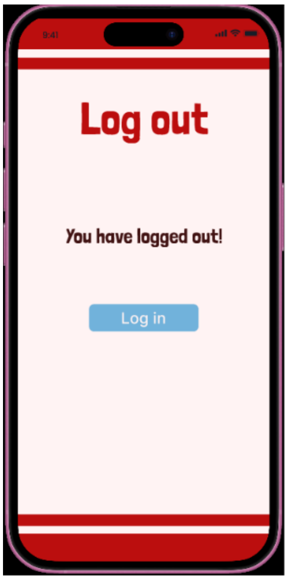


Figure 31: Post-log out page

UI Component	Input Type	Purpose	Validation
“Log in” button	Command button	Redirects user to user log in	None

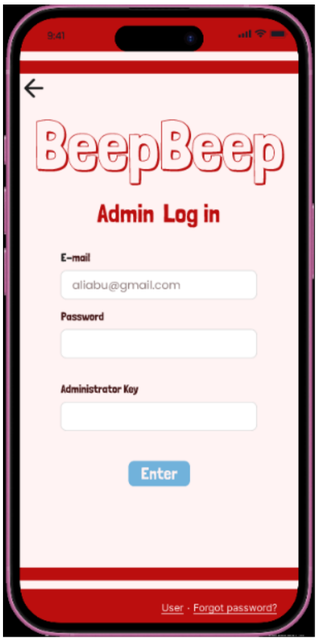


Figure 32: Administrator log in page

UI Component	Input Type	Purpose	Validation
Email field	Text field	Captures the user’s registered email	Must match a valid email; cannot be left blank

Password field	Text field	Captures the secure password associated with the account	Requires a minimum of 8 characters
Administrator Key field	Text field	Captures a secure administrator key	Requires a minimum of 8 characters
Log in “Enter” button	Command button	Submits the E-mail and Password input to authenticate the user.	Disabled if input fields are empty; triggers an error message if the account credentials do not match database records.
Register “Enter” button	Command button	Redirects new users to the sign-up/registration form screen.	None; immediately clears current inputs and shifts the viewport to the Registration page.
User Link	Hyperlink	Redirects users to the dedicated user portal.	Routes users to the user log in/registration interface.
Forgot Password? Link	Hyperlink	Directs users to a recovery page if they cannot remember their credentials.	Triggers a secure verification/OTP layout to the user's email.

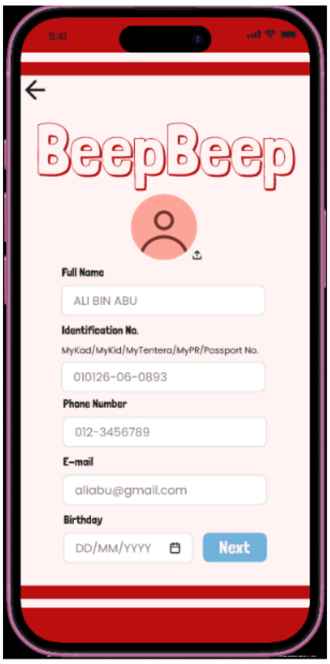


Figure 33: Registration page

UI Component	Input Type	Purpose	Validation
Upload Button	Command Button	Allows user to upload an image as a profile picture for customisation purposes.	None
Full Name, Identification No., E-mail and Phone Number Field	Text Field	Captures the user's related personal information	None
Birthdate	Calendar Picker Modal	Captures the user's birth date.	Restricts future dates and dates more recent than 10 years ago.
"Next" Button	Command Button	Submits user's personal information; directs user to the Nickname page	None
Go back arrow	Command Button	Redirects user to the previous page, log in/registration page.	None
Bottom Navigation Bar	Tab Bar (Icons)	Allows global navigation between History/Bookings, Home, and Profile (current view) settings.	Instantly shifts the app layout context to the selected administrative or user-specific screen.

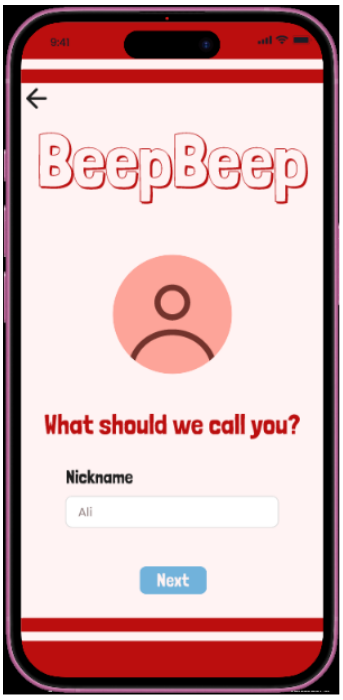


Figure 34: Nickname page

UI Component	Input Type	Purpose	Validation
Nickname field	Text field	Captures user's desired nickname	None
Next Button	Text Field	Submits user's nickname; directs user to Welcome page.	None

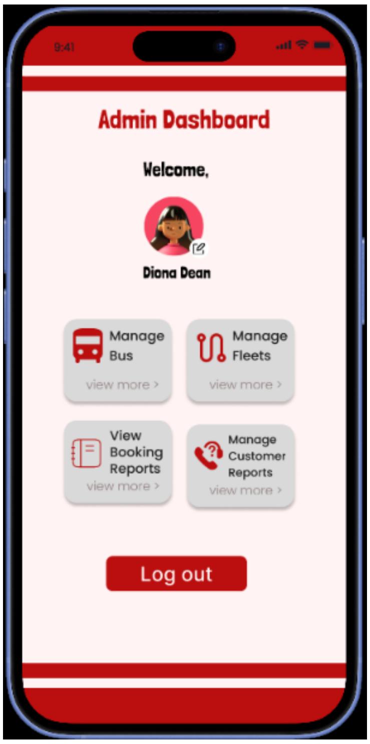


Figure 35: Admin Dashboard interface

UI Component	Input Type	Purpose	Validation
Upload Button	Command Button	Allows user to upload an image as a profile picture for customisation purposes.	None
"Manage Bus" Card	Grid Navigation Button	Directs the administrator to the management screen for individual buses.	None
"Manage Fleets" Card	Grid Navigation Button	Directs the administrator to the fleets management page.	None

“View Booking Reports” Card	Grid Navigation Button	Directs the administrator to the analytics panel	None
“Manage Customer Reports” Card	Grid Navigation Button	Directs the administrator to the customer report page	None
“Log out” button	Command Button	Directs user to the Log Out Confirmation page	None

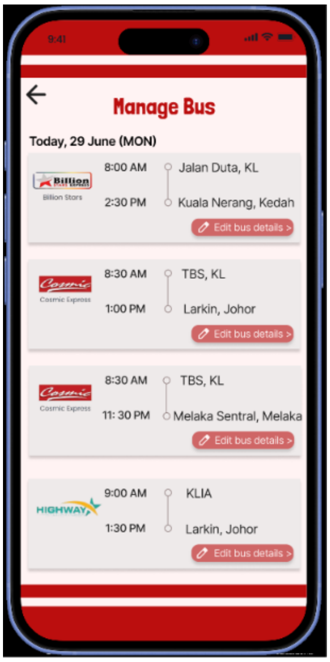


Figure 36: Manage Bus page

UI Component	Input Type	Purpose	Validation
Back Arrow Button	Command Button	Redirects the administrator back to the previous screen	None
Edit bus details Button	Grid Item Action Button	Launches the contextual editing portal for that specific scheduled trip block	Disabled if the selected trip has already departed or concluded
Trip Log Card List	Scrollable Feed	Houses and arranges individual schedule blocks chronologically according to departure timestamps.	Bound by data constraints

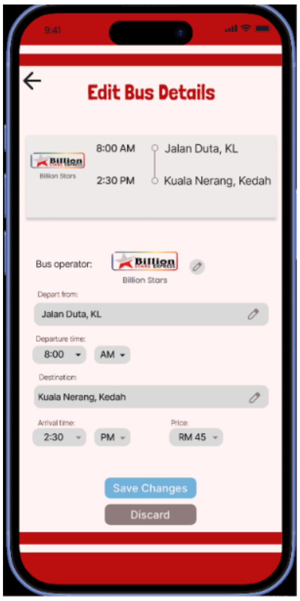


Figure 37: Edit Bus Details page

UI Component	Input Type	Purpose	Validation
Depart from:	Text field	Captures the user's desired terminal departure location	Restricts input to valid terminal locations served by the company network to eliminate typing errors
Destination:	Text field	Captures the user's final arrival station	Cannot be identical to the "Depart from" selection; dynamically lists available connecting hubs
Bus operator Edit Icon	Image Selection	Captures updates to the transit operator company entity	Restricts selection to valid, registered bus logistics operators stored within the partner database framework.
Departure and Arrival time Pickers	Dropdown Menu Selectors	Captures the specific hourly/minute time and period designation for schedule launching	Time must follow valid 12-hour formatting structures
Price Selector	Numeric Dropdown Menu	Captures the structural pricing value assigned per ticket seat assignment	Capped between predefined corporate minimum/maximum price values to

			avoid severe data-entry errors.
"Save changes" Button	Command Action Button	Commits all modified route schedule variables permanently to the backend logistics database.	Kept disabled until at least one parameter string has been updated and passes validation checks
“Discard” Button	Command Action Button	Reverts all temporary form changes back to the pre-existing database record states.	Erases active input cache values instantly and navigates the admin back out to the main Management List.

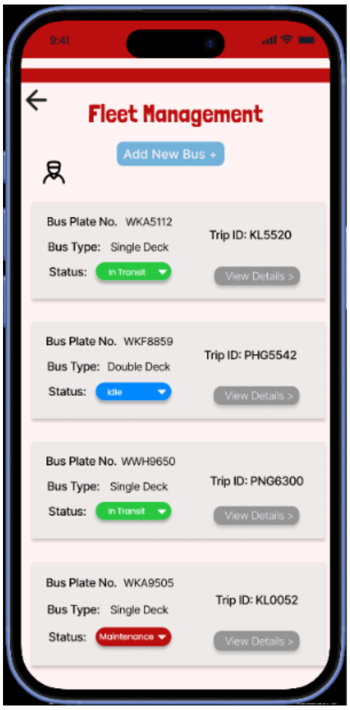


Figure 38: Fleet management page

UI Component	Input Type	Purpose	Validation
“Add New Bus” Button	Command Navigation Button	Redirects the administrator to a blank registry form to add a newly acquired vehicle to the database.	None
Status selector	Drop down menu	Captures instant changes to a bus’s active status state	Restricts input values to a closed system list
Go back arrow	Command Button	Redirects user to the previous page, the admin dashboard page.	None
“View details” Button	Command button	Launches the full logistical summary, active passenger manifesto, and telemetry data for a specific Trip ID	None

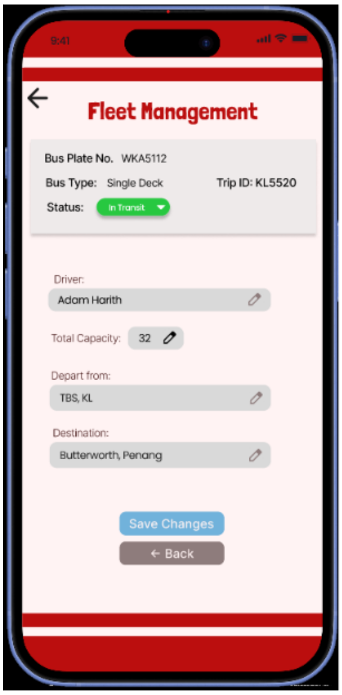


Figure 39: Add new bus interface

UI Component	Input Type	Purpose	Validation
"Save Changes" Button	Command Button	Transmits and commits the modified configuration array to the centralized fleet database framework.	Disabled by default until at least one parameter string has been updated
"Back" Button	Action Navigation Button	Dismisses the current modification matrix and safely returns the admin to the primary list dashboard.	Clears the active input field memory buffer without saving uncommitted changes to prevent data pollution.
Driver:	Text field	Captures or assigns the full name of the employee driving the coach	Must map to an active, licensed driver profile within the staff roster database
Total capacity	Text Box	Captures the absolute number of available passenger seats on the bus frame	Restricts admin from entering any value below 20.
Depart from:	Text field	Captures adjustments to the registered route departure platform or terminal	Constrained to valid terminal nodes within the corporate operating network system

Destination:	Text field	Captures adjustments to the final arrival station or terminal location	Relies on relational constraints
Status selector	Drop down menu	Captures the instant high-level logistical status of the vehicle	Constrained to valid predefined state tags
Go back arrow	Command Button	Redirects user to the previous page, the admin dashboard page.	None

Figure 40: Booking Reports page

UI Component	Input Type	Purpose	Validation
Go back arrow	Command Button	Redirects user to the previous page, the admin dashboard page.	None

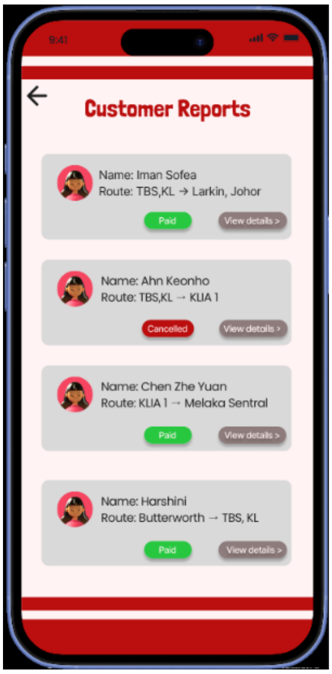


Figure 41: Customer Reports page

UI Component	Input Type	Purpose	Validation
Go back arrow	Command Button	Redirects the administrator to the previous page, the admin dashboard page.	None
“View details” Button	Command Button	Directs the administrator to customer report panel	None

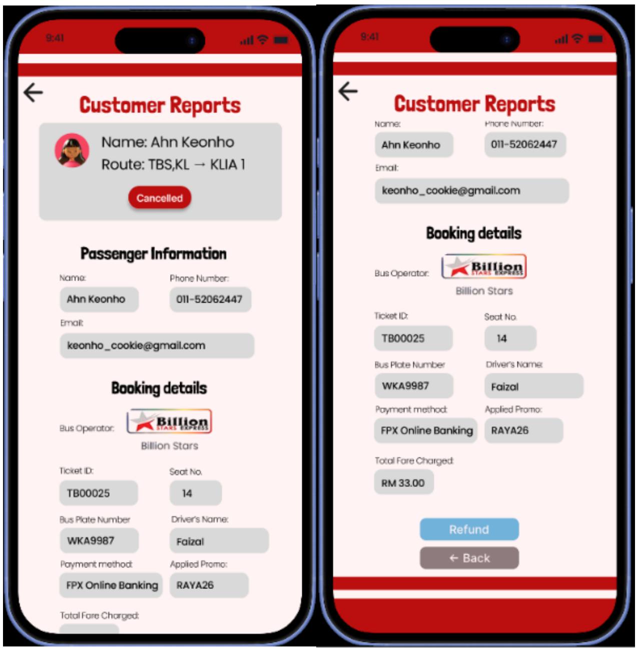


Figure 42 and Figure 43: Individual customer report interface

UI Component	**Input Type**	**Purpose**	**Validation**
Go back arrow	Command Button	Redirects user to the previous page, which is the main Customer Report page.	None
“Refund” Button	Command Action Button	Redirects user to the Refund Confirmation page	Disabled if the customer did not request a refund
“Back” Button	Command Button	Redirects user to the previous page, which is the main Customer Report page.	None

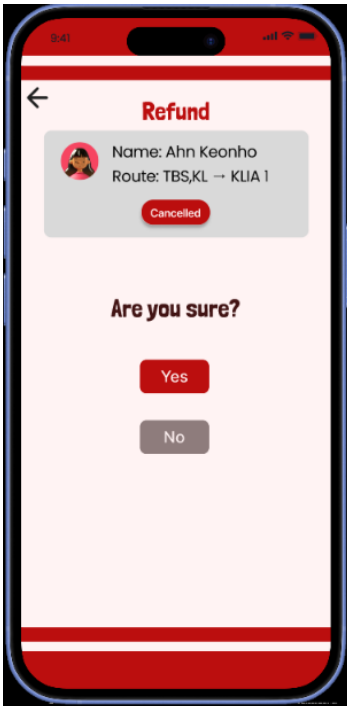


Figure 44: Refund confirmation page

UI Component	**Input Type**	**Purpose**	**Validation**
“Yes” button	Command action button	Confirms manual refund	None
“No” button	Command button	Cancels refund action	None
Go back arrow	Command Button	Redirects user to the previous page, which is the individual customer report page.	None

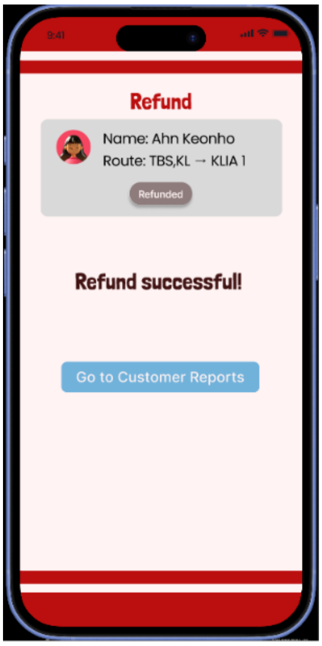


Figure 45: Post-refund page

UI Component	Input Type	Purpose	Validation
"Go to Customer Reports" Button	Command button	Redirects user to the main Customer Report page.	None

9.0 Summary of the Proposed System

The proposed system, **BeepBeep**, is a comprehensive, mobile-first bus transit and fleet management ecosystem designed to completely modernize traditional, manual ticketing workflows. By shifting operations away from fragmented paper records and whiteboard scheduling, the system unifies all facets of passenger logistics and administrative control into a centralized digital platform. The architecture is strategically partitioned into two primary frameworks: an accessible, frictionless interface for public passengers and a robust, high-security backend portal for operators and administrative personnel.

For public users, the system offers an intuitive ticket procurement stream that significantly optimizes the booking experience. Customers can dynamically filter and query available bus schedules by utilizing structured input fields, toggling journey types between one-way and round-trip routes, adjusting passenger counts, and selecting verified terminal locations. Furthermore, the application provides users with comprehensive account autonomy via a dedicated profile center. Within this space, passengers can manage communication preferences through interactive notification controls, review historical travel logs, and configure advanced multi-layered transaction security elements, such as biometric hardware integration and customized security PIN codes.

Administratively, the platform functions as an operational command center that dramatically increases management efficiency and data integrity. Transitioning staff to a digital environment is safeguarded by strict authentication layers; standard user accounts can only be elevated to administrative roles by satisfying multi-factor authentication codes alongside verified corporate security keys. Once authorized, the administrative backend automates data processing pipelines to deliver real-time output designs. Managers are equipped with instant business intelligence analytics, glanceable performance widgets tracking key performance indicators—such as total revenue, ticket sales volume, and fleet occupancy rates—and interactive financial trend visualizations, complete with on-demand reporting utilities to support swift operational decision-making.

10.0 Appendices

- Link to prototype: Figma Prototype